



PRODUCT PRESENTATION

AgriShield

Offline-first parametric micro-insurance for smallholder farmers

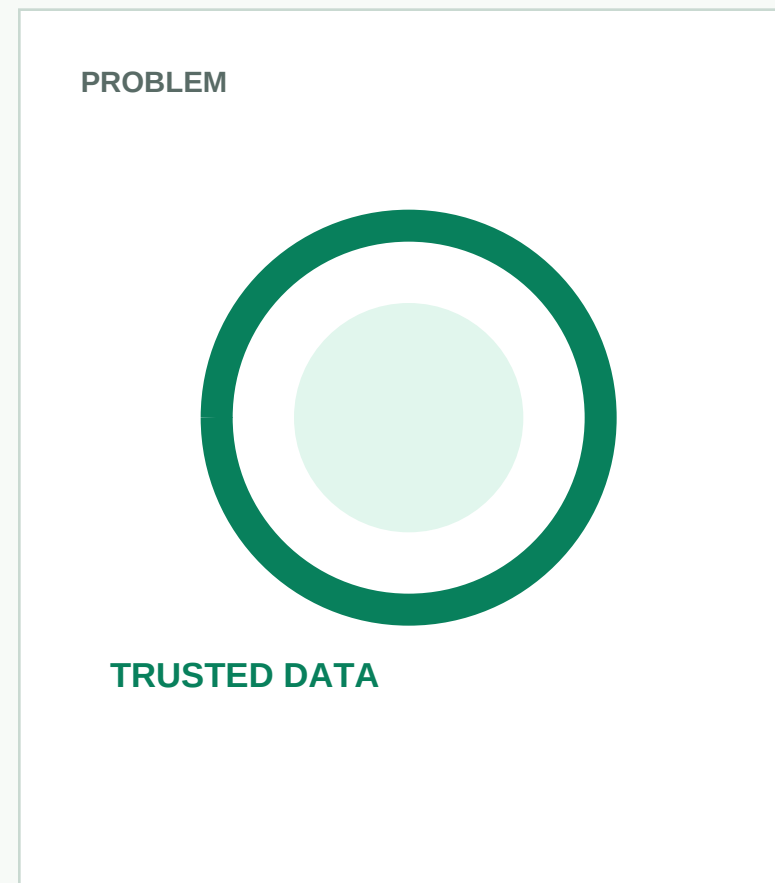


Weather-triggered protection that keeps working when connectivity does not.

The protection gap is also a connectivity gap

- Smallholder farmers face rainfall volatility, delayed claims, and limited access to trusted insurance.
- Rural connectivity can be intermittent exactly when a farmer needs to submit evidence.
- Manual verification slows payouts and makes every claim expensive to process.

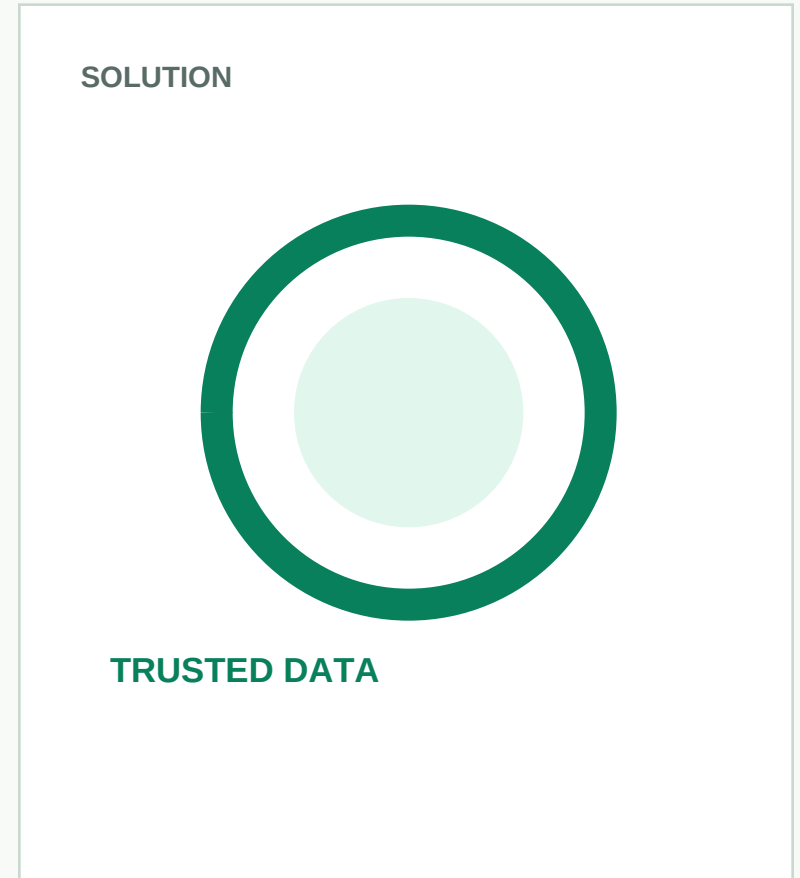
Design principle: make the claim journey simple, auditable, and resilient offline.



A parametric safety net built around verified weather

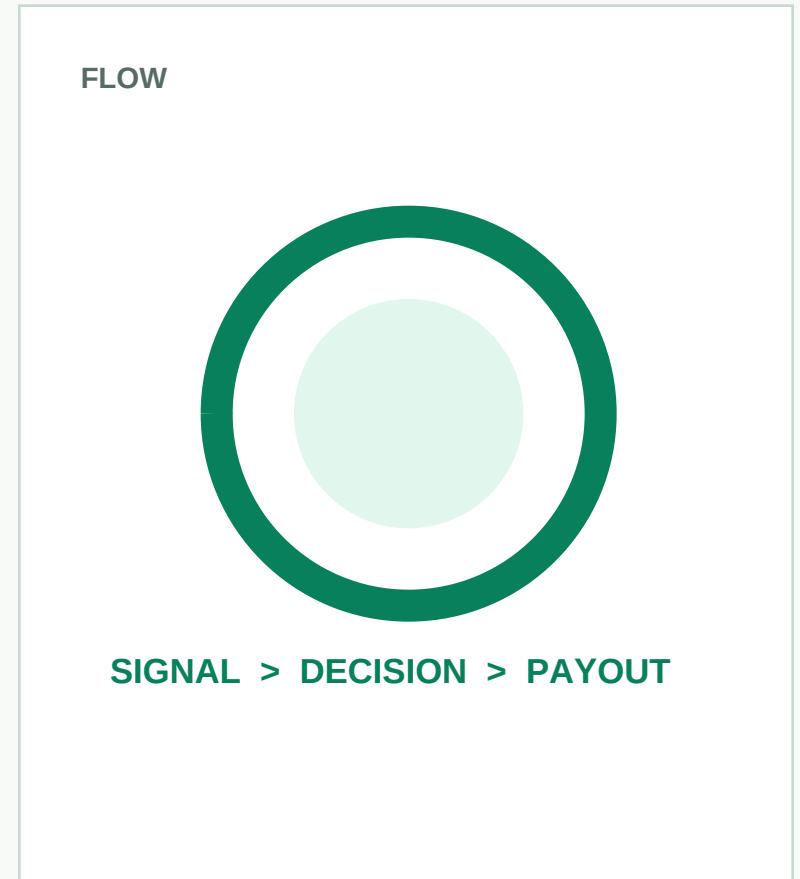
- Policies define a crop, location, coverage period, rainfall trigger, and payout amount.
- Three weather sources are combined using a median consensus, with outliers flagged.
- Eligible claims move through verification and payout workflows without manual field inspection.

Outcome: transparent rules, faster decisions, and predictable support.



From rainfall signal to farmer payout

- 1. Weather readings arrive from three independent sources.
- 2. The policy engine calculates a verified rainfall median.
- 3. The farmer submits a claim online or saves it locally while offline.
- 4. The backend verifies the trigger, records an audit trail, and initiates payout.

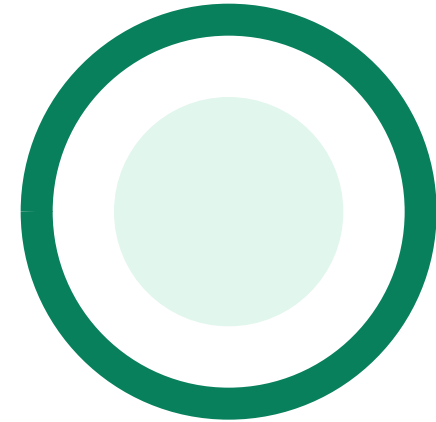


Architecture: small, focused, and deployable

- React and TypeScript frontend with responsive farmer and administrator views.
- Node.js and Express REST API for authentication, policies, weather, claims, payouts, and audit logs.
- SQLite with WAL mode keeps the deployment lightweight and self-contained.

The same codebase supports development, production static hosting, and offline web use.

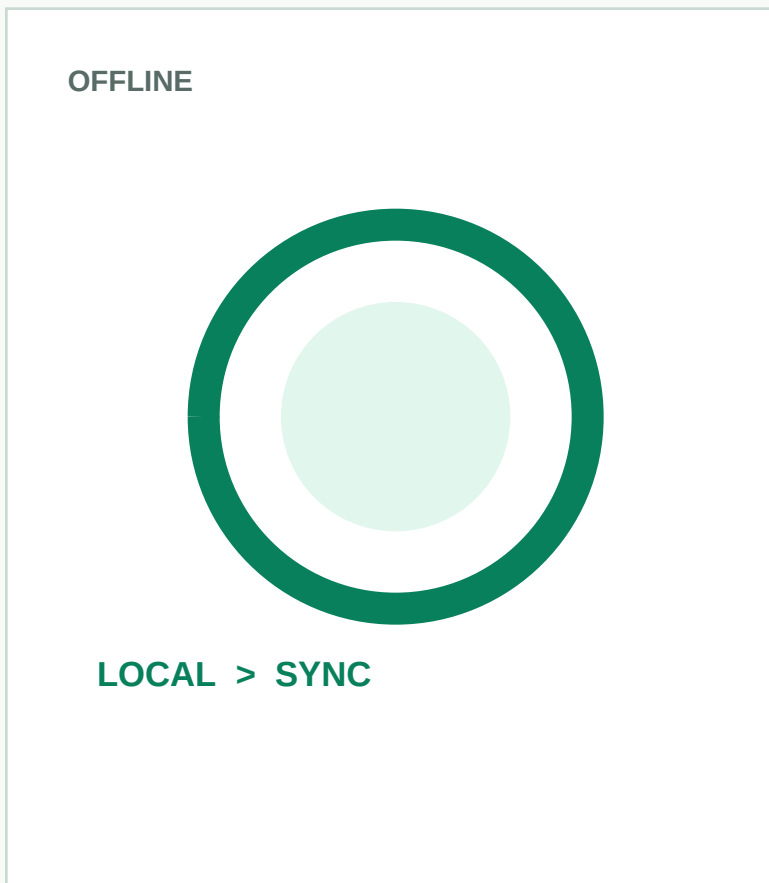
ARCHITECTURE



UI > API > DATA

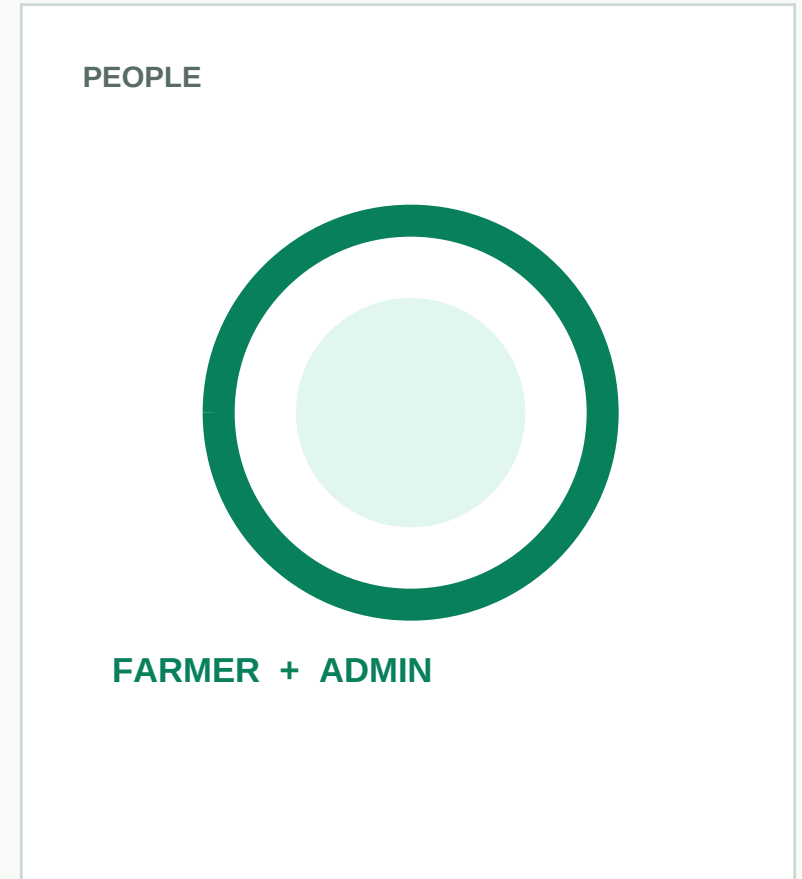
Offline-first is a product behavior, not a fallback screen

- Claims are stored locally when the network is unavailable.
- A visible queue shows what is waiting to sync.
- Reconnection triggers synchronization and preserves the user journey.
- Cached profiles and weather context keep the experience useful between connections.



Two experiences, one trusted source of truth

- Farmer: view policy, rainfall status, claim history, payout decision, and voice explanation.
- Administrator: monitor devices, inspect claims, review weather consensus, settle payouts, and audit actions.
- English and Telugu language support meets users where they are.





AGRICULTURE / INSURANCE / TRUST

Make insurance reachable when it matters most

AgriShield turns weather data into a simple, auditable promise: verify fairly, pay quickly, keep working offline.



Next steps: connect production weather providers, integrate regulated payout rails, and pilot with local farmer groups.